CS 340 README Project Two

About the Project/Project Title

This project is a full-stack dashboard application that was developed for the company, Grazioso Salvare. This company is an international search and rescue training organization. This dashboard allows its users to filter the data in order to visualize the data coming from shelters in Austin, Texas. It can be used to identify the most ideal candidates for training based on a few components (breed, age, etc.)

Motivation

The motivation for this program is to allow users to easily filter data in order to find the most ideal candidates for water rescue, mountain/wilderness rescue, and disaster recovery.
The extra features in this app, such as the shelter's location on a map and the chart showing the distribution of different dog breeds, allow the user to easily find the information necessary and save time in the long run due to the program identifying the best candidates in seconds.

Getting Started

In order to replicate this project, the following would need to be done:
-Clone the project from the repository on GitHub
-Ensure Python is installed
-Install the required packages for Python
-Run the ProjectTwoDashboard.ipynb
-Select a row from the table in order to view the chart paired with a map

Installation

The tools needed for this project would be the following:
-Python (can be installed from their website)
-Jupyter would need to be installed if the user isn't using a tool like codio
-pymongo
-dash

Usage

The program can be utilized by running the app and selecting to open it from the link it spits out. Once the dashboard is accessed, the radio buttons can be selected in order to filter the data by the ideal candidates for a specific rescue type, like water rescue or disaster. The interactive table updates based on the filter selected and if the circle is checked on a particular animal, the graph will update with their location. The pie chart updates following the filtering of the data with the breed distribution for that type of rescue.

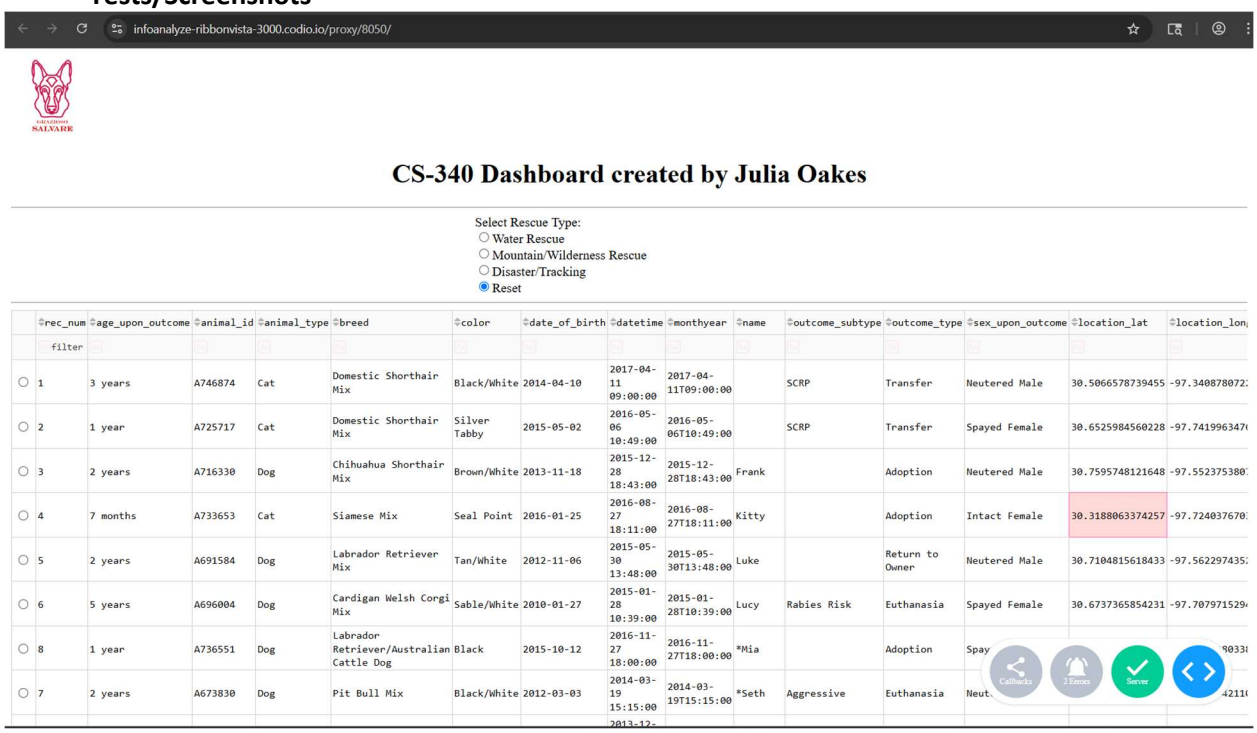
Code Example

```
@app.callback(Output('datatable-id','data'),
              [Input('filter-type', 'value')])
def update_dashboard(filter_type):
    if filter_type == 'water':
        query = {
            "breed": {"$in": ["Labrador Retriever Mix", "Chesapeake Bay Retriever", "Newfoundland"]},
            "age_upon_outcome_in_weeks": {"$lte": 104}
        }
    elif filter_type == 'mountain':
```

Note: This template has been adapted from the following sample templates: Make a README, Best README Template, and A Beginners Guide to Writing a Kickass README.

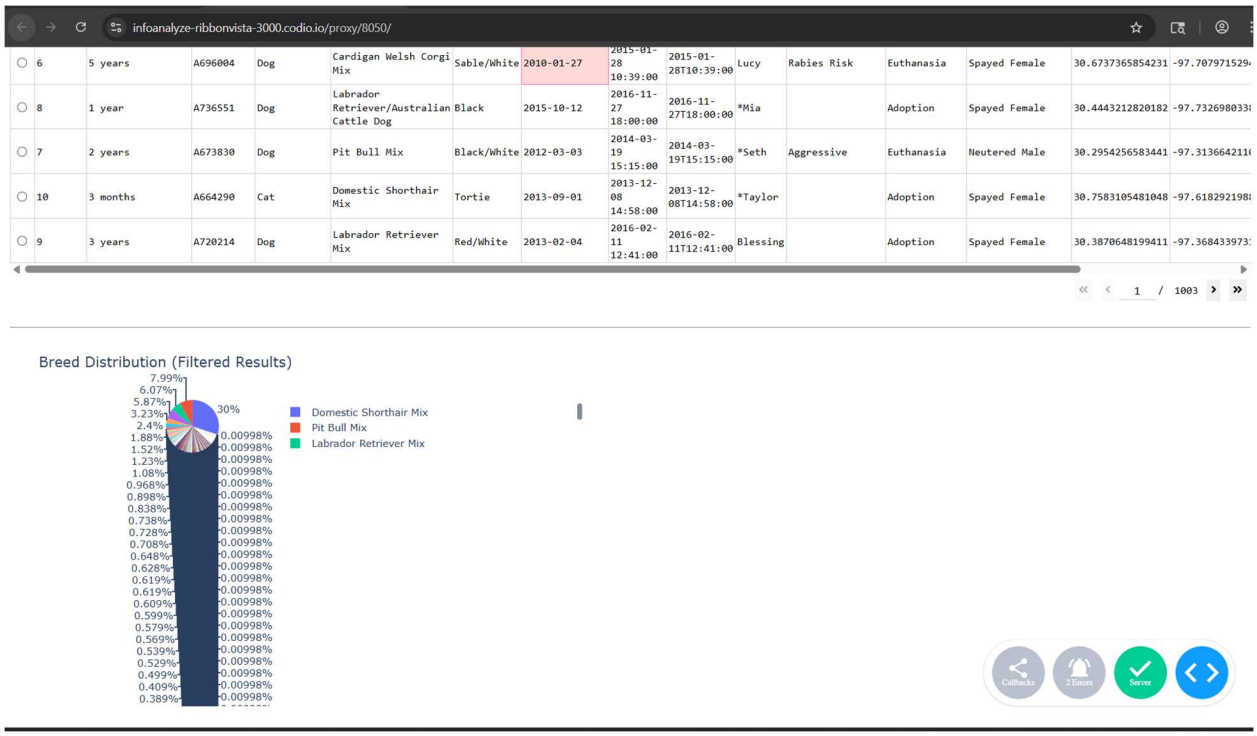
```
        query = {
            "breed": {"$in": ["German Shepherd", "Alaskan Malamute", "Old English Sheepdog",
"Siberian Husky", "Rottweiler"]}
        }
    elif filter_type == 'disaster':
        query = {
            "breed": {"$in": ["Doberman Pinscher", "German Shepherd", "Golden Retriever",
"Bloodhound", "Rottweiler"]}
        }
    else:
        query = {}

    result = db.read(query)
    filtered_df = pd.DataFrame.from_records(result)
    if '_id' in filtered_df.columns:
        filtered_df.drop(columns=['_id'], inplace=True)

    return filtered_df.to_dict('records')
```

Tests/Screenshots

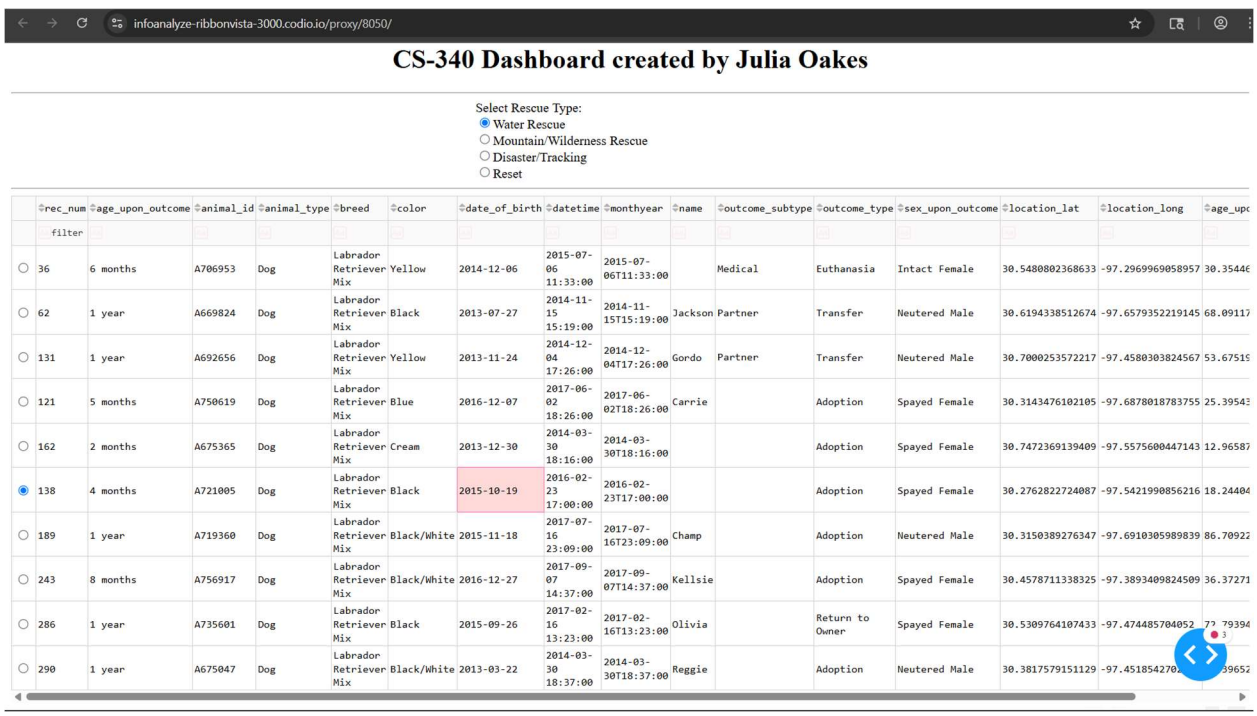


	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome	location_lat	location_lon
	filter														
	1	3 years	A746874	Cat	Domestic Shorthair Mix	Black/White	2014-04-10	2017-04-11 09:00:00	2017-04-11T09:00:00		SCRP	Transfer	Neutered Male	30.5066578739455	-97.340878072
	2	1 year	A725717	Cat	Domestic Shorthair Mix	Silver Tabby	2015-05-02	2016-05-06 10:49:00	2016-05-06T10:49:00		SCRP	Transfer	Spayed Female	30.6525984560228	-97.741996347
	3	2 years	A716330	Dog	Chihuahua Shorthair Mix	Brown/White	2013-11-18	2015-12-28 18:43:00	2015-12-28T18:43:00	Frank		Adoption	Neutered Male	30.7595748121648	-97.552375380
	4	7 months	A733653	Cat	Siamese Mix	Seal Point	2016-01-25	2016-08-27 18:11:00	2016-08-27T18:11:00	Kitty		Adoption	Intact Female	30.3188063374257	-97.724037670
	5	2 years	A691584	Dog	Labrador Retriever Mix	Tan/White	2012-11-06	2015-05-30 13:48:00	2015-05-30T13:48:00	Luke		Return to Owner	Neutered Male	30.7104815618433	-97.562297435
	6	5 years	A696004	Dog	Cardigan Welsh Corgi Mix	Sable/White	2010-01-27	2015-01-28 10:39:00	2015-01-28T10:39:00	Lucy	Rabies Risk	Euthanasia	Spayed Female	30.6737365854231	-97.707971529
	8	1 year	A736551	Dog	Labrador Retriever/Australian Cattle Dog	Black	2015-10-12	2016-11-27 18:00:00	2016-11-27T18:00:00	*Mia		Adoption	Spay		9033
	7	2 years	A673830	Dog	Pit Bull Mix	Black/White	2012-03-03	2014-03-19 15:15:00	2014-03-19T15:15:00	*Seth	Aggressive	Euthanasia	Neut		4211
								2013-12-							

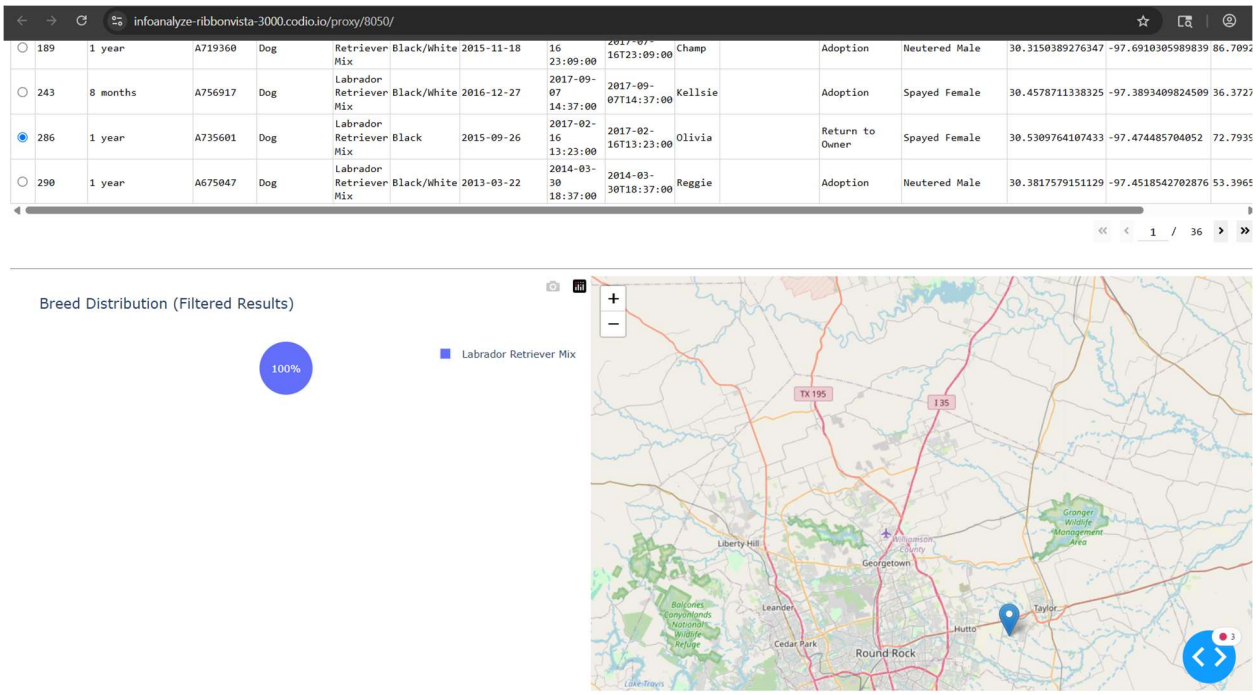
Note: This template has been adapted from the following sample templates: [Make a README](), [Best README Template](), and [A Beginners Guide to Writing a Kickass README]().



*This is with no filters selected



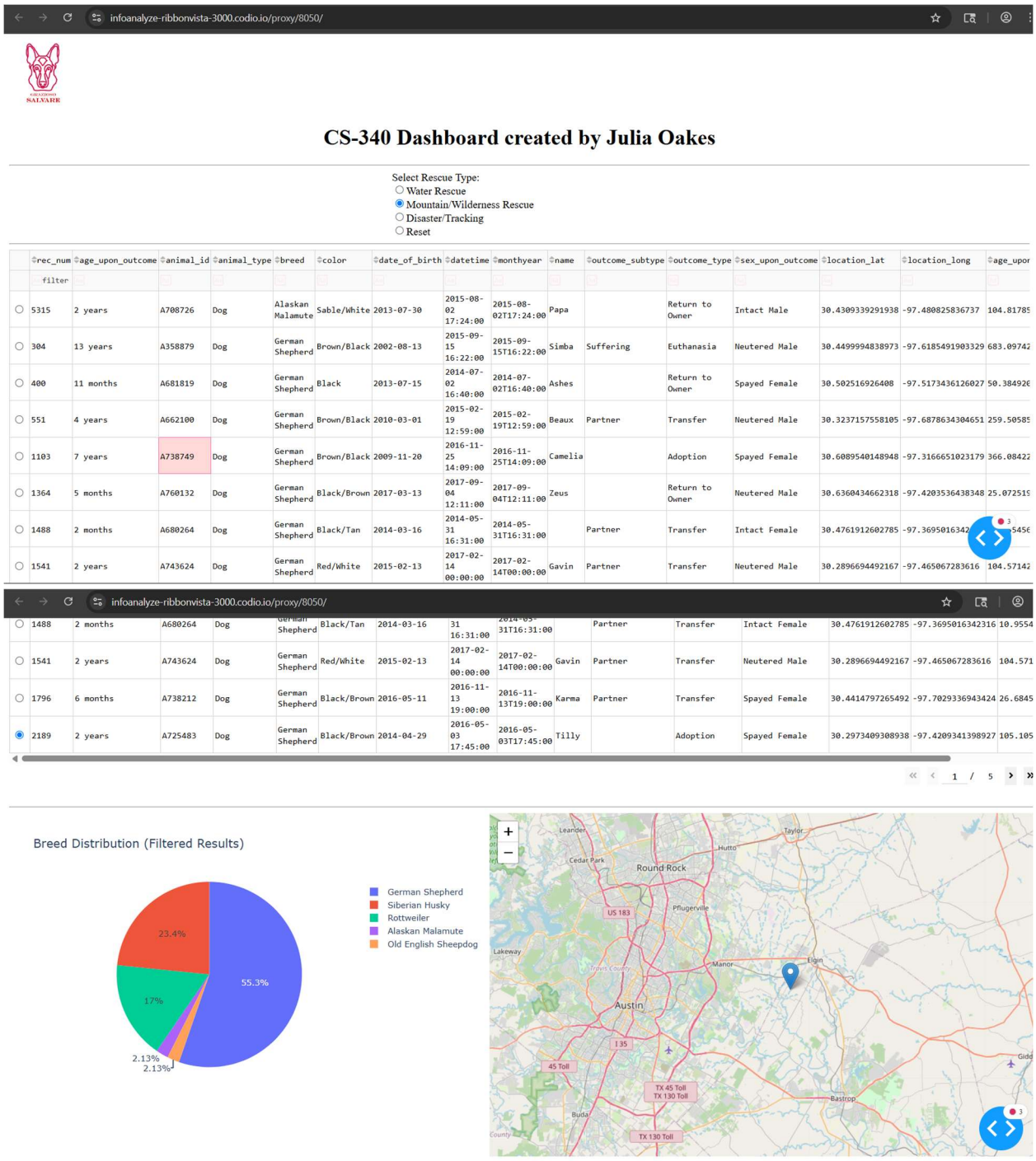
Note: This template has been adapted from the following sample templates: [Make a README](), [Best README Template](), and [A Beginners Guide to Writing a Kickass README]().



*This was with “Water Rescue” selected as well as a specific dog chosen to demonstrate the map functionality.

Note: This template has been adapted from the following sample templates: [Make a README](), [Best README Template](), and [A Beginners Guide to Writing a Kickass README]().

Southern New Hampshire University



rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome	location_lat	location_long	age_upon
5315	2 years	A708726	Dog	Alaskan Malamute	Sable/White	2013-07-30	2015-08-02 17:24:00	2015-08-02T17:24:00	Papa		Return to Owner	Intact Male	30.4309339291938	-97.480825836737	104.81785
304	13 years	A358879	Dog	German Shepherd	Brown/Black	2002-08-13	2015-09-15 16:22:00	2015-09-15T16:22:00	Simba	Suffering	Euthanasia	Neutered Male	30.4499994838973	-97.6185491903329	683.09742
400	11 months	A681819	Dog	German Shepherd	Black	2013-07-15	2014-07-02 16:40:00	2014-07-02T16:40:00	Ashes		Return to Owner	Spayed Female	30.502516926408	-97.5173436126027	50.384926
551	4 years	A662100	Dog	German Shepherd	Brown/Black	2010-03-01	2015-02-19 12:59:00	2015-02-19T12:59:00	Beaux	Partner	Transfer	Neutered Male	30.3237157558105	-97.6878634304651	259.50585
1103	7 years	A738749	Dog	German Shepherd	Brown/Black	2009-11-20	2016-11-25 14:09:00	2016-11-25T14:09:00	Camelia		Adoption	Spayed Female	30.6089540148948	-97.3166651023179	366.08422
1364	5 months	A760132	Dog	German Shepherd	Black/Brown	2017-03-13	2017-09-04 12:11:00	2017-09-04T12:11:00	Zeus		Return to Owner	Neutered Male	30.6360434662318	-97.4203536438348	25.072519
1488	2 months	A680264	Dog	German Shepherd	Black/Tan	2014-03-16	2014-05-31 16:31:00	2014-05-31T16:31:00		Partner	Transfer	Intact Female	30.4761912602785	-97.3695016342	5456
1541	2 years	A743624	Dog	German Shepherd	Red/White	2015-02-13	2017-02-14 00:00:00	2017-02-14T00:00:00	Gavin	Partner	Transfer	Neutered Male	30.2896694492167	-97.465067283616	104.57142
1796	6 months	A738212	Dog	German Shepherd	Black/Brown	2016-05-11	2016-11-13 19:00:00	2016-11-13T19:00:00	Karma	Partner	Transfer	Spayed Female	30.4414797265492	-97.7029336943424	26.6845
2189	2 years	A725483	Dog	German Shepherd	Black/Brown	2014-04-29	2016-05-03 17:45:00	2016-05-03T17:45:00	Tilly		Adoption	Spayed Female	30.2973409308938	-97.4209341398927	105.105

*This is with Mountain/Wilderness Rescue selected

Note: This template has been adapted from the following sample templates: [Make a README](), [Best README Template](), and [A Beginners Guide to Writing a Kickass README]().

Southern New Hampshire University

infoanalyze-ribbonvista-3000.codio.io/proxy/8050/

CS-340 Dashboard created by Julia Oakes

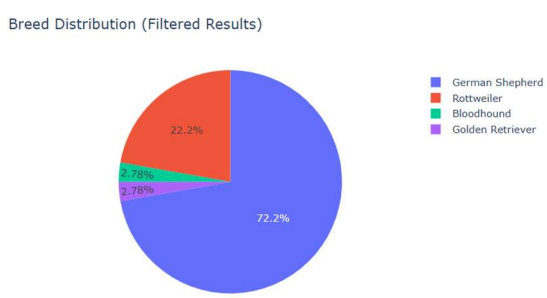
Select Rescue Type:
- ○ Water Rescue
- ○ Mountain/Wilderness Rescue
- ◉ Disaster/Tracking
- ○ Reset

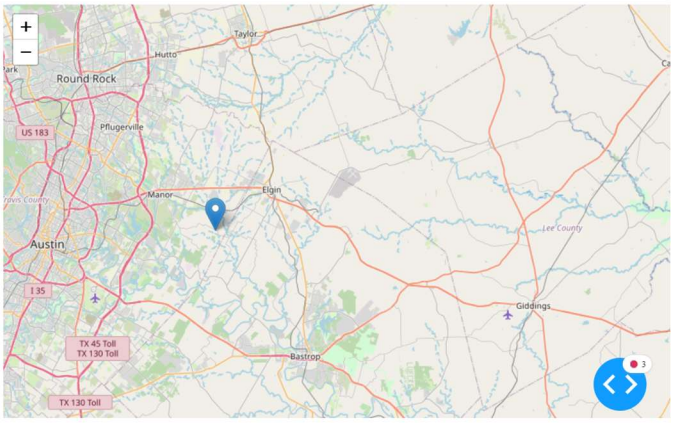
	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome	location_lat	location_long	age_u
	filter															
○	3767	4 years	A712291	Dog	Bloodhound	Red	2011-09-20	2015-09-22 15:43:00	2015-09-22T15:43:00	Boomer		Return to Owner	Intact Male	30.2709983761287	-97.5923916912722	209.09
○	304	13 years	A358879	Dog	German Shepherd	Brown/Black	2002-08-13	2015-09-15 16:22:00	2015-09-15T16:22:00	Simba	Suffering	Euthanasia	Neutered Male	30.4499994838973	-97.6185491903329	683.09
○	400	11 months	A681819	Dog	German Shepherd	Black	2013-07-15	2014-07-02 16:40:00	2014-07-02T16:40:00	Ashes		Return to Owner	Spayed Female	30.502516926408	-97.5173436126027	50.384
○	551	4 years	A662100	Dog	German Shepherd	Brown/Black	2010-03-01	2015-02-19 12:59:00	2015-02-19T12:59:00	Beaux	Partner	Transfer	Neutered Male	30.3237157558105	-97.6878634304651	259.50
○	1103	7 years	A738749	Dog	German Shepherd	Brown/Black	2009-11-20	2016-11-25 14:09:00	2016-11-25T14:09:00	Camelia		Adoption	Spayed Female	30.6089540148948	-97.3166651023179	366.08
○	1364	5 months	A760132	Dog	German Shepherd	Black/Brown	2017-03-13	2017-09-04 12:11:00	2017-09-04T12:11:00	Zeus		Return to Owner	Neutered Male	30.6360434662318	-97.4203536438348	25.072
○	1488	2 months	A680264	Dog	German Shepherd	Black/Tan	2014-03-16	2014-05-31 16:31:00	2014-05-31T16:31:00		Partner	Transfer	Intact Female	30.4761912602785	-97.36950163	955
○	1541	2 years	A743624	Dog	German Shepherd	Red/White	2015-02-13	2017-02-14 00:00:00	2017-02-14T00:00:00	Gavin	Partner	Transfer	Neutered Male	30.2896694492167	-97.465067283616	104.57

infoanalyze-ribbonvista-3000.codio.io/proxy/8050/

○	1488	2 months	A680264	Dog	German Shepherd	Black/Tan	2014-03-16	31 16:31:00	2014-05-31T16:31:00		Partner	Transfer	Intact Female	30.4761912602785	-97.3695016342316	10.9554
◉	1541	2 years	A743624	Dog	German Shepherd	Red/White	2015-02-13	2017-02-14 00:00:00	2017-02-14T00:00:00	Gavin	Partner	Transfer	Neutered Male	30.2896694492167	-97.465067283616	104.571
○	1796	6 months	A738212	Dog	German Shepherd	Black/Brown	2016-05-11	2016-11-13 19:00:00	2016-11-13T19:00:00	Karma	Partner	Transfer	Spayed Female	30.4414797265492	-97.7029336943424	26.6845
○	2189	2 years	A725483	Dog	German Shepherd	Black/Brown	2014-04-29	2016-05-03 17:45:00	2016-05-03T17:45:00	Tilly		Adoption	Spayed Female	30.2973409308938	-97.4209341398927	105.105

« ‹ 1 / 4 › »





*This is with Disaster/Tracking Selected

Roadmap/Features (Optional)

Note: This template has been adapted from the following sample templates: [Make a README](), [Best README Template](), and [A Beginners Guide to Writing a Kickass README]().

Some potential future features that could be implemented would be the following:
-The ability to mark multiple dogs at the same time so their locations on the map can be viewed in tandem.
-A bar chart could be added for additional information to be displayed, like the ages of the dogs

Contact
Julia Oakes
Julia.oakes@snhu.com

Note: This template has been adapted from the following sample templates: Make a README, Best README Template, and A Beginners Guide to Writing a Kickass README.